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Hoggan et al.

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(54) **ZERO SIGN-ON AUTHENTICATION**

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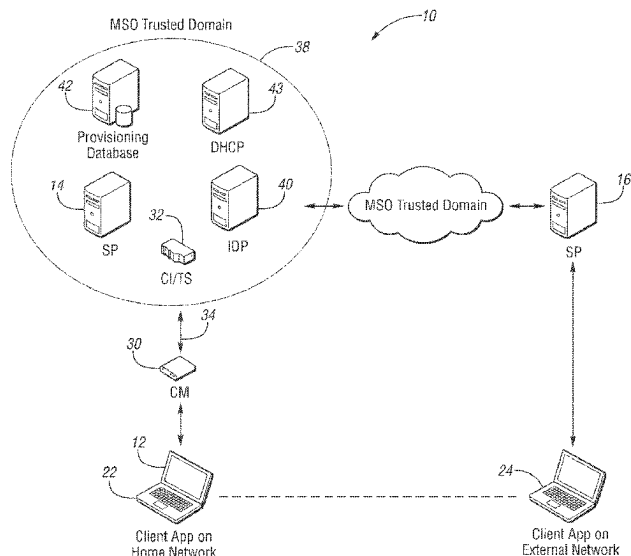
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(57) **ABSTRACT**

A method of facilitating zero sign-on access to media
services depending on trust credentials. The trust credentials
may be cookies, certificates, and other data sets operable to
be stored on a device used to access the media services such
that information included therein may be used to control the
zero sign-on capabilities of the user device.

18 Claims, 6 Drawing Sheets



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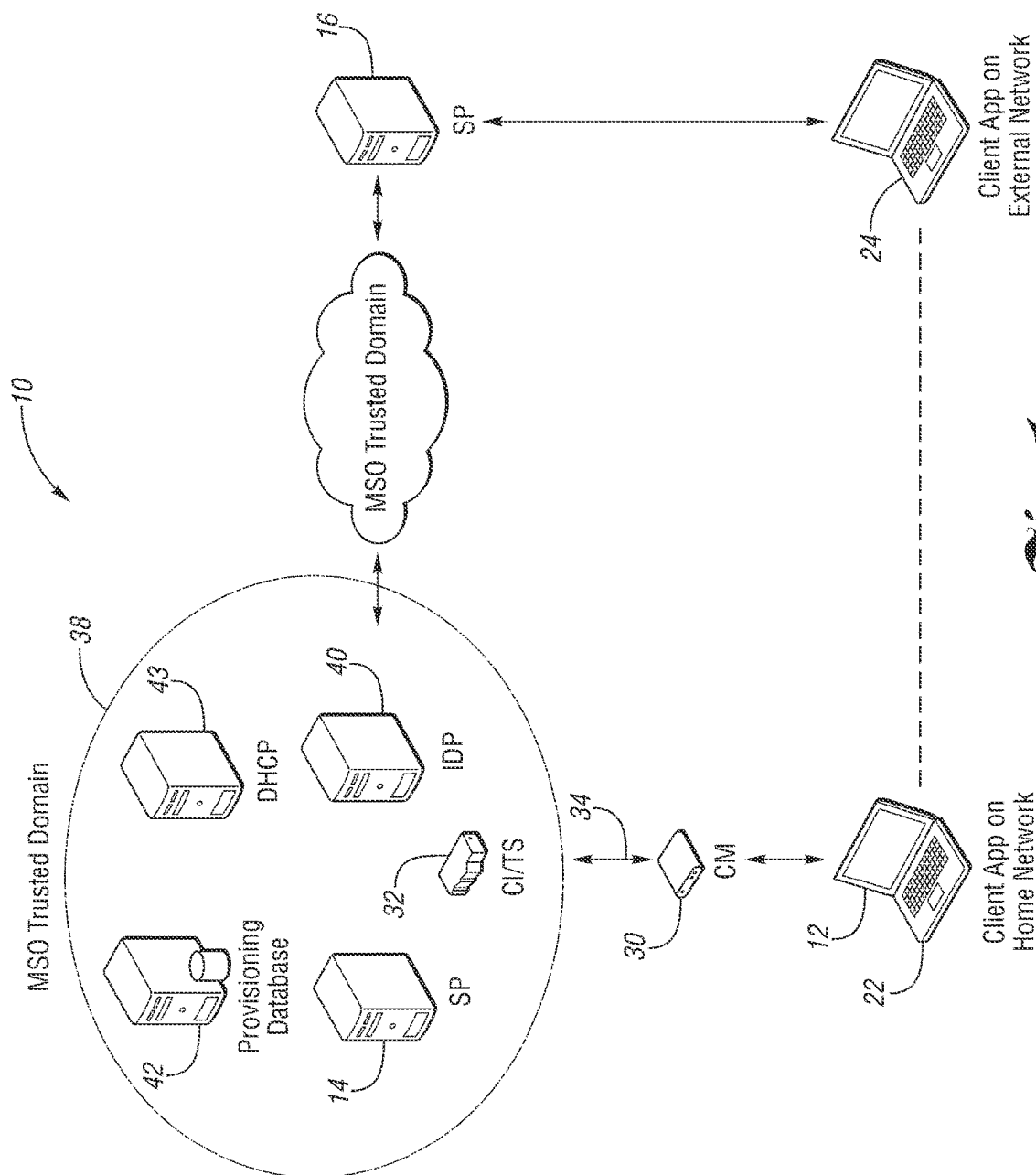


Fig. 1

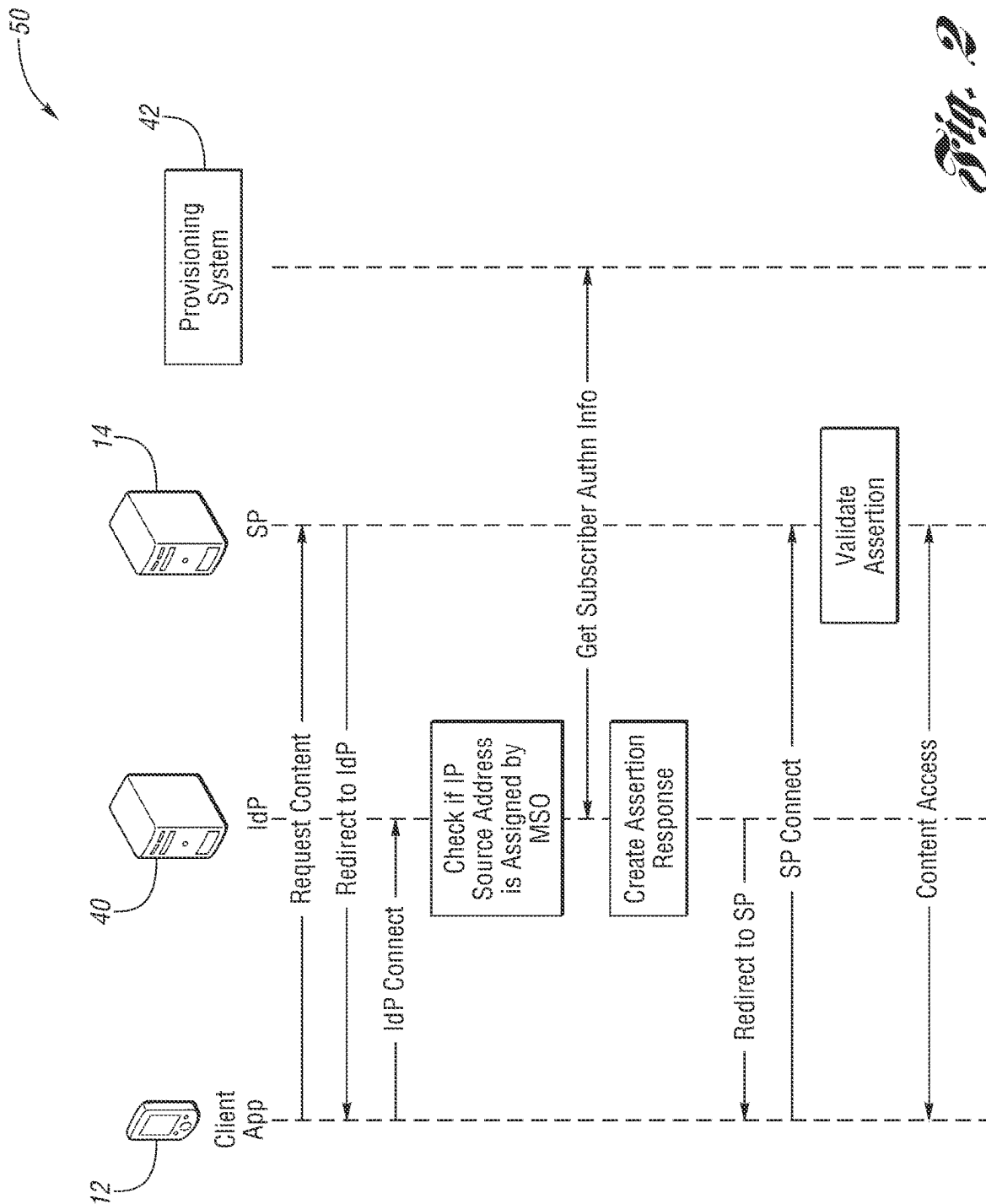


Fig. 2

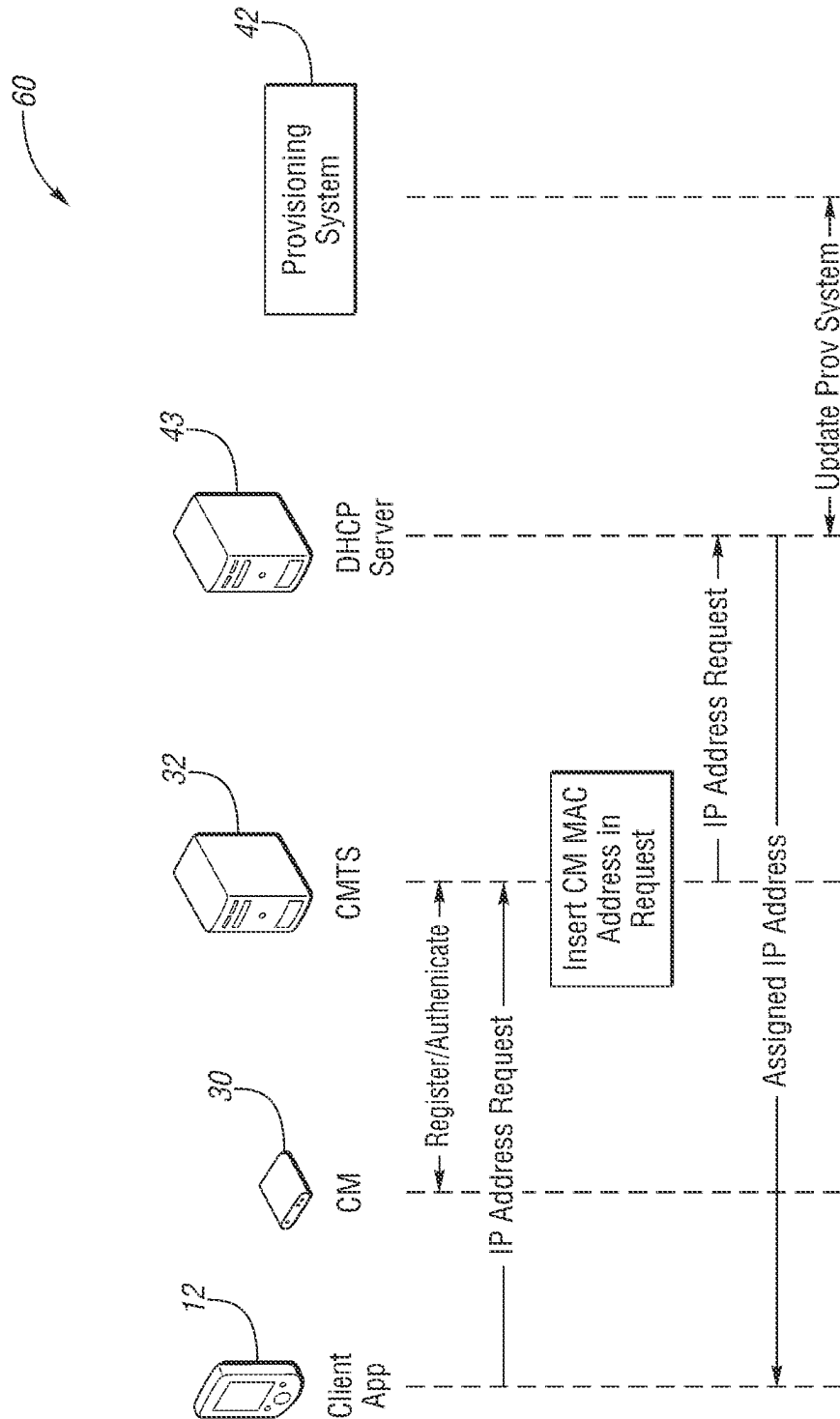
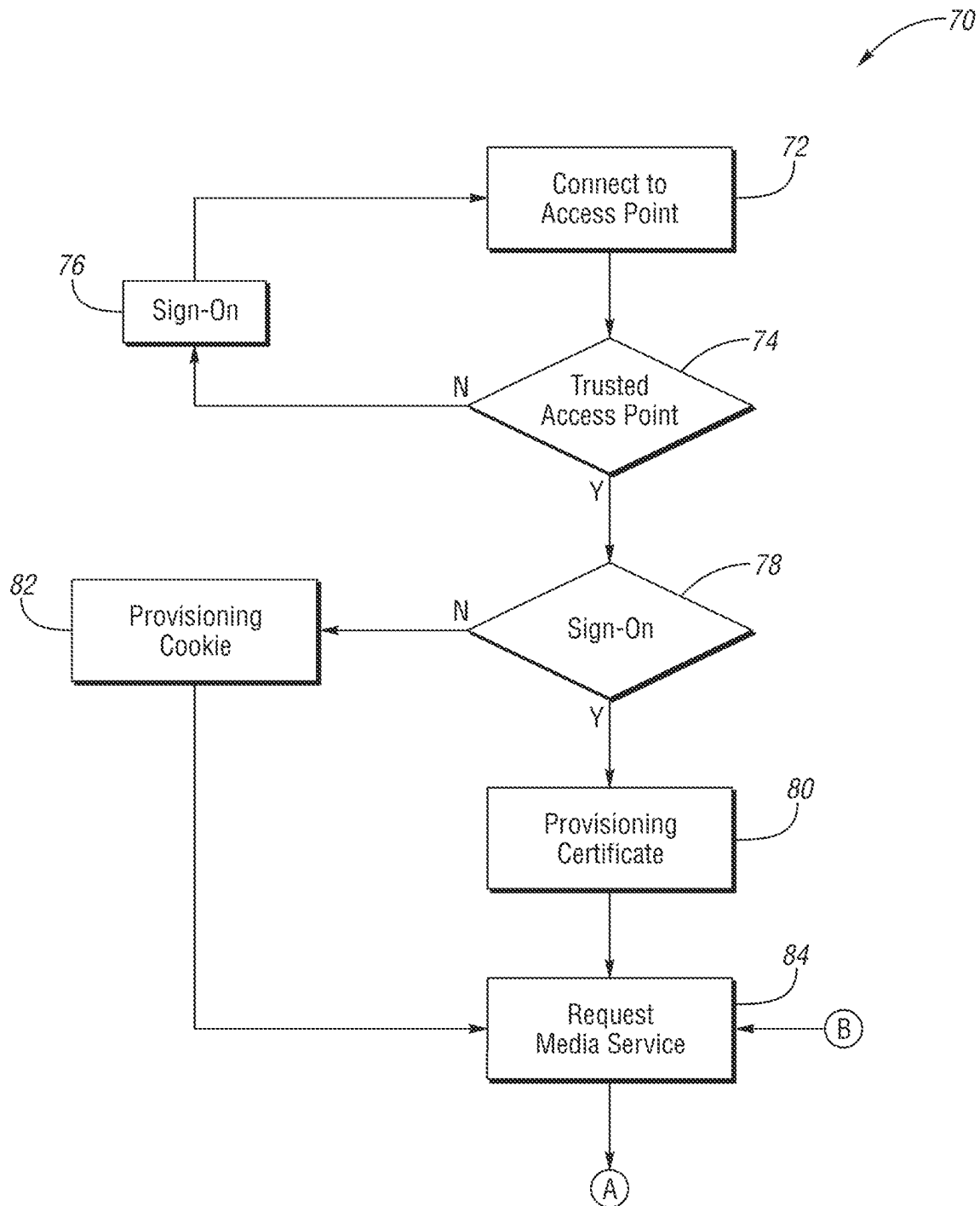


Fig. 3

*Fig. 4A.*

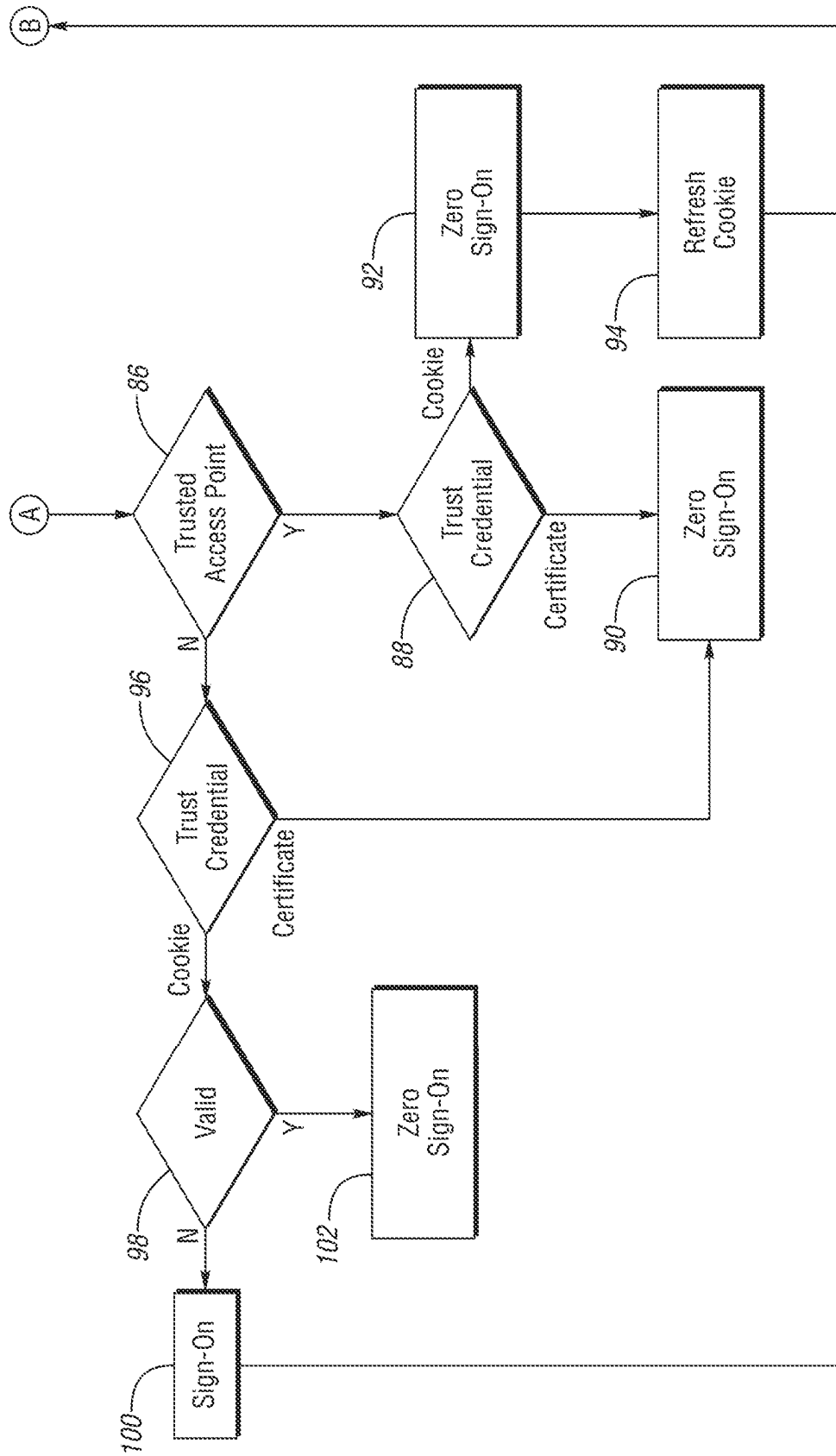
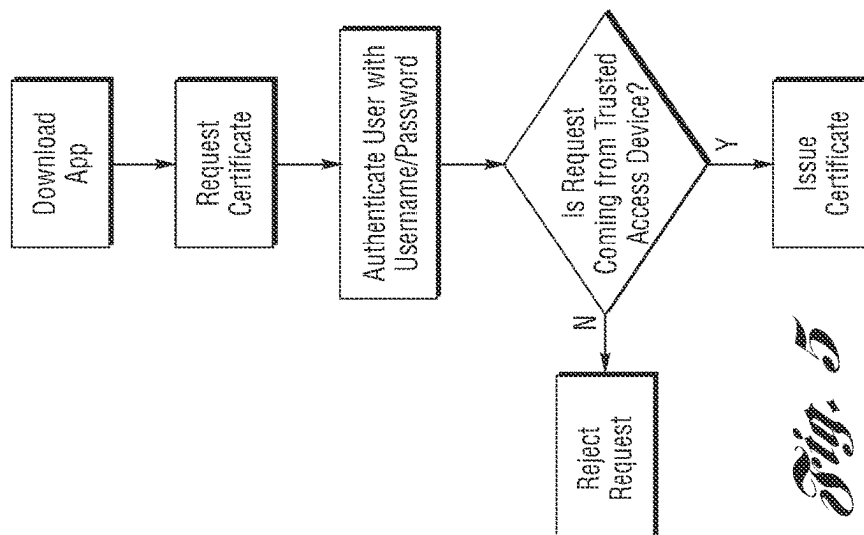
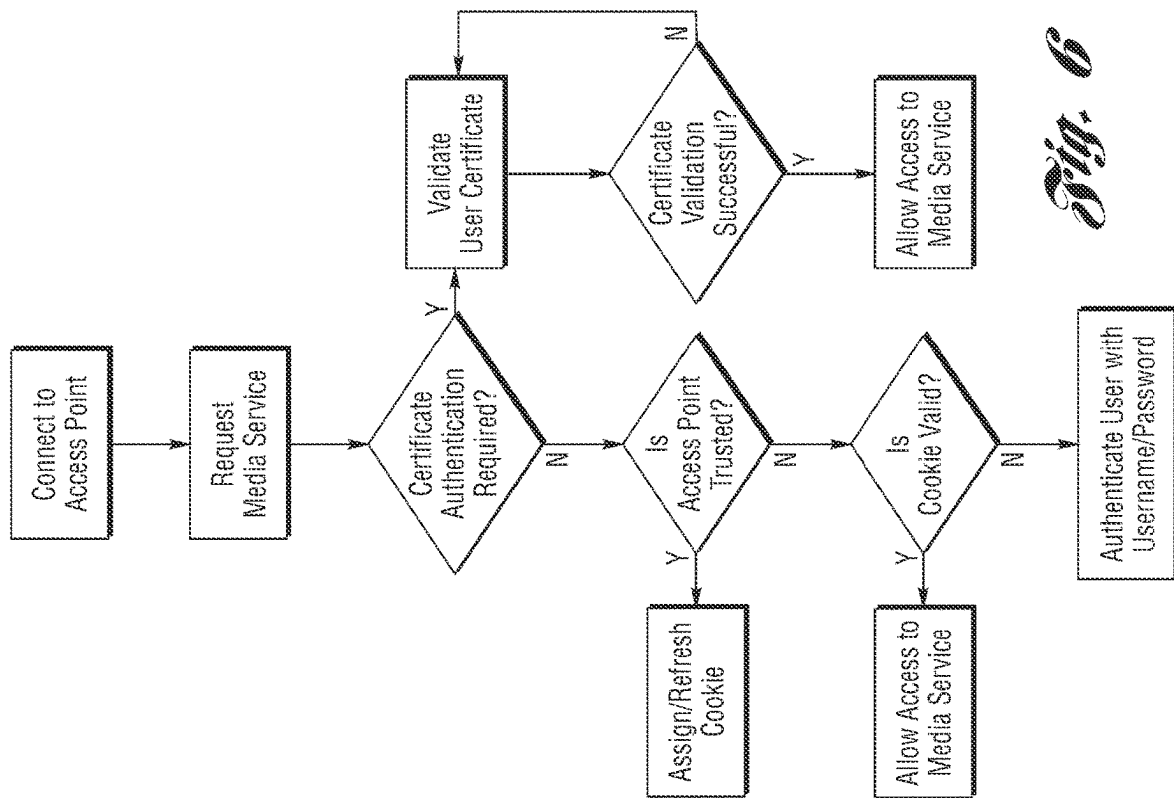


Fig. 4B



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ZERO SIGN-ON AUTHENTICATION**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 14/617,757, filed Feb. 9, 2015, which application is a continuation of U.S. patent application Ser. No. 13/173,630, filed Jun. 30, 2011, the disclosures and benefits of which are hereby incorporated in their entireties by reference herein.

TECHNICAL FIELD

The present invention relates to facilitating user access to media services without requiring a user to sign-on or enter a password as part of an authentication process required before permitting access to the media services.

BACKGROUND

People who use the Internet to buy goods and services typically have to establish an identity at each business, which may require the customer to complete a sign-on process of entering a username and password each time a business's website is accessed or any other time services are requested. As more and more businesses rely on online services, customers may be required to keep track of numerous username and password combinations, i.e., one for each website, which can become a burden on the users. There is also the risk of identity theft of the user's username and passwords since that information may be stored in so many different places. Accordingly, a need exists to limit the number of sign-on operations a user must perform and/or the number of locations at which the user's username and password combinations are stored.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is pointed out with particularity in the appended claims. However, other features of the present invention will become more apparent and the present invention will be best understood by referring to the following detailed description in conjunction with the accompany drawings in which:

FIG. 1 illustrates a zero sign-on system contemplated by one non-limiting aspect of the present invention.

FIG. 2 illustrates a messaging diagram of zero sign-on messaging contemplated by one non-limiting aspect of the present invention.

FIG. 3 illustrates a messaging diagram of address allocation messaging contemplated by one non-limiting aspect of the present invention.

FIGS. 4a-4b illustrate a flowchart of a method for supporting a zero sign-on process contemplated by one non-limiting aspect of the present invention.

FIG. 5 illustrates a flowchart of certificate provisioning as contemplated by one non-limiting aspect of the present invention.

FIG. 6 illustrates a flowchart of a method for supporting a zero sign-on process contemplated by one non-limiting aspect of the present invention.

DETAILED DESCRIPTION

As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that

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the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

FIG. 1 illustrates a zero sign-on system 10 contemplated by one non-limiting aspect of the present invention. The system 10 supports zero sign-on access to any type of suitable media service for a number of user devices 12. For exemplary purposes, a single mobile user device 12, e.g., a mobile computer, phone, tablet, PDA, etc., is shown to be accessing IP based services through signaling exchanged with first and second service providers (SPs) 14, 16. The user device 12 may be any other device, including but not limited to a settop box (STB), media terminal adapter (MTA), and media player. The user device 12 may include a client application embodied in a computer-readable medium having stored thereon a plurality of instructions including instructions which, when executed by a processor, cause the processor to facilitate the operation contemplated by the present invention to support zero sign-on.

The client application may be configured to provide a user/subscriber interface and to otherwise support interactions with the service provider of multiple system operator (MSO) 14. It could be a common web browser, such as Internet Explorer or Firefox, or it could be a custom application or any corresponding type of operating system. The client application can be configured to operate on any type of device 12 such as a PC, laptop, or smart phone. A first position 22 of the user device 12 is shown to correspond with the home or other location of a user that subscribes to services of the first service provider 14, which may be the home of a user associated with the user device or the home of another individual that happens to subscribe to services from the same first service provider 14. A second position 24 corresponds with the user device 12 changing locations to another location, such as a Wi-Fi hotspot, where the second service provider 16 is required to support at least some of the signaling necessary for the user device to access services of the first service provider 14.

At the first location 12, the user device 12 is shown to be connected to a cable modem (CM) 30 to facilitate data exchange with a Cable Modem Termination System (CMTS) 32 of the service provider. The cable modem 30 and the CMTS 32 may cooperate to enable Data Over Cable Service Interface Specification (DOCSIS) compliant bi-directional high speed data access across a cable network 34. The cable modem 30 may be located in a Subscriber's home and may be embedded in a VoIP device (eMTA) or set top. The CMTS 32 may be a central manager associated with the service provider 14 to control network access. A trusted domain 38 of the service provider 14 generally corresponds with multiple CMTSs 32, and the cable modems 30 they support, and other devices 40, 42, 43 with which it manages to facilitate interacting media services with its subscribers.

The second service provider 16 is not shown to include a trusted domain, however, it may similarly include such a trusted domain. The trust domain of the second service provider 16 is not shown since the user device 12 may not be a subscriber to the second service provider 16, and thereby, an untrusted user device. The service providers 14, 16 may be configured to simultaneously support communications between trusted and untrusted user devices, or more

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specifically, between access points used by those devices to communicate with the service providers. The cable modem **30** and/or CMTS **32** acts as the access point when the user device **12** is at the first position **22** and the wireless router or other gateway (not shown) acts as the access point when the user device is at the second location **24**. Since the access point at the second location **24** is not within the trusted domain of the first service provider **14**, it is considered as an untrusted access point with respect to the first service provider **14** (it may be a trusted access point with respect to the second service provider **16**).

The present invention, while predominately described with respect to supporting access to media services offered by satellite, broadcast, and cable television service providers; high speed data service providers; and telephony service providers, fully contemplates its use with any type of media service that relies on signal communications from a service provider, and optionally, where the service provider is able to communicate the signals to different locations and to different types of user devices. One exemplary illustration of providing television services is set forth to demonstrate one useful aspect of the present invention, namely, supporting IP streaming of television programs to user devices **12** regardless of the location of the user device **12** and the access point (e.g., trusted and untrusted) used to facilitate communication of the related signaling. The television services may include subscriber specific services, such as personal video recording (PVR), premium channels, etc., and non-subscriber specific services, such as broadcast programming, public access, etc.

The streaming of the television signals may be supported with the CMTS **32** in the case of a cable based communication system or some other analogous devices in the case of satellite, high speed data, and telephony based service provider system configurations. The service provider **14** may include a server, headend unit, or other device to support and source television programs, previously recorded video (VOD, PVR, etc.), and other media content (video games, webpages, etc.) to the CMTS **32** for subsequent packaging and transmission to one or more access points (gateways). U.S. application Ser. No. 12/650,664 filed Dec. 31, 2009, the disclosure of which is incorporated in its entirety by reference herein, discloses a similar system of facilitating zero sign-on based on a trust state of cable modems or other network access parts configured to support message communications between the CMTS and the local user devices.

A provisioning database **42** may be associated with the first service provider and included in the trusted domain to facilitate regulation of the media services available to and/or consumed by the user device **12**. The provisioning database **42** may contain information about a level of service or types of services paid for by a subscriber. It may also maintain a list of IP addresses that have been assigned to client applications associated with the cable modem **30**, e.g., if the subscriber allows multiple user devices to connect to its cable modem, each of those user devices may be assigned a unique IP address within the subnet of the corresponding CMTS **32**. Optionally, each of the assigned IP addresses may be bound with a MAC address assigned to the supporting cable modem **30** in order to facilitate associating IP addresses with a particular cable modem **30** and/or subscriber.

While the first location **22** is shown to support DOCSIS based communications, the present invention is not intended to be so limited and fully contemplates the service providers **14**, **16** being configured to support virtually any other type

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of communications, including cellular or other wireline/wireless phone related communications. The service providers **14**, **16** may be configured to facilitate access to virtually any type of electronic media or media services/content. When a subscriber wants to access content they can command a client application to request it from the service provider **14**, **16**. Before the service provider **14**, **16** allows access to content, either through a trusted or untrusted access point, it may be required to authenticate the user device **12** and/or the subscriber thereof, and then check if they are authorized for the requested access.

FIG. 2 illustrates a messaging diagram **50** of the messaging that may be exchanged to facilitate the zero sign-on contemplated by one non-limiting aspect of the present invention. To authenticate a request from a user device **12**, the service provider **14** may send an authentication request in the client application redirect to an identity provider (IdP) **40** and wait for a response that contains an assertion indicating that the subscriber has been authenticated. The response may also contain some authorization information. The IdP **40** may be a centralized subscriber authentication entity within a trusted domain **38** of the service provider **14** and/or with a trusted third party (not shown). When requested by the service provider **14**, the IdP **40** may be able to authenticate subscribers and provide an assertion token to the service provider **14** that can be used for making content access decisions. The IdP **40** may be configured with the IP subnet used by a DHCP server **43** to assign IP addresses to the client application within the MSO's trusted domain/network.

If the service provider **14** does not know the IdP **40** of the subscriber it may prompt the subscriber for it. If the service provider needs additional authorization data it may directly issue requests to the MSO's authorization server (not shown) after the subscriber has been authenticated. The service provider **14** can be located anywhere, e.g., at the first and second locations **22**, **24**. It can be located in the back-end or on a gateway device (IPS) of an MSO's network. It can also be located on an external web site. Wherever the client application connects to request services/content is where the service provider **14** may be considered to be located.

One non-limiting aspect of the present invention contemplates enabling zero sign-on for a subscriber to access services and content at different web sites (MSO and Vendor) without having to enter a username and password to login. It does this by leveraging the existing security infrastructure of the cable network or other type of network associated with the service provider **14**. When the client application requests content from the service provider **14** it may be redirected to the IdP **40** for authentication. When the IdP **40** receives the redirected authentication request from the client application it may check to see if the IP source address (e.g., IP address assigned by the access point) is within the MSO's IP address subnet for client devices **12** (e.g., within its trusted domain **38**).

If the IP source address is within the trusted domain **38**, the IdP **40** queries the provisioning system **42** for subscriber information ID using the IP source address. The IdP **40** may then create the assertion response and send it to the service provider **14**. Importantly, this authentication may be done without prompting the subscriber for authentication credentials (username and password). After verifying the assertion token the service provider **14** may allow the client application to access content. If the redirected authentication request has not come from the MSO's network the IdP **40**

would require the subscriber to authenticate using their credentials (username password) unless the Client App has a valid security context.

After the IdP 40 has authenticated the client application using zero sign-on it may establish an authentication session or security context with the client application and create a temporary token or cookies that is/are stored on the client application. When a service provider (any SP) redirects the client application again to the IdP 40 for authentication, the IdP 40 can immediately respond with an authentication assertion if the cookie is still valid (not expired) without having to prompt the subscriber. Optionally, as long as the client application has a valid authentication session cookie it can connect to any service provider 14, 16 that is in the same circle of trust as the IdP 40 and not have to require its subscriber to login with credentials. This is independent of the access network used, which means the subscriber can move their client application from the cable network to another access network, connect to a service provider 16 in a different domain and continue to still enjoy zero sign-on functions.

The IdP 40 uses the IP source address of the client application to query the provisioning system 42 for subscriber info in order to create an appropriate assertion response. FIG. 3 illustrates a messaging diagram 60 for determining the subscriber associated with the IP source address in accordance with one non-limiting aspect of the present invention. When a client application requests an IP address the CMTS 32 inserts the subscriber's cable modem 30 MAC address, which is known and trusted by the service provider 14, in the request before forwarding it to a DHCP server 43. After the DHCP server 43 assigns an IP address to the client application, the DHCP server 43 may then notify the provisioning system 42 and indicate which cable modem 30 MAC address it is associated with. It may also notify the provisioning system 42 when an IP address is not associated with a given cable modem MAC address (expires or assigned to another device). Therefore, the provisioning system may create a mapping of client application IP source addresses that are associated with a subscriber's cable modem MAC address.

Once the cookie expires, the client application would need to be re-authenticated, but that can be done automatically (zero sign-on) when the client application is connected to the cable network again. If the client application was not able to connect to the cable network before the cookie expires, the IdP 40 can prompt the subscriber for credentials (username and password) which would renew the authentication session token (single sign-on). The cookie may be provisioned to last for a certain period, such as one day, week, month, six months, before it needs to be refreshed. The cookie may be used in this manner to facilitate zero sign-on access to websites or other service provider portals used by subscribers to access media for the period without the subscriber having to sign-on. Since the cookie may be stored on the computer-readable medium of the user device 12, any user in possession of the user device 12 would receive access to the corresponding media services.

One non-limiting aspect of the present invention contemplates enhancing the security credentials (e.g., token, cookie, etc.) stored on the user device 12 by instead storing a certificate. The certificate may be more robust than username and password credentials. Passwords can be too simple and easily guessed by hackers or vulnerable to dictionary attacks. They can also be shared with friends leading to unauthorized access to service. Digital certificates are not subject to these vulnerabilities and provide a stronger

level of subscriber authentication. Once a subscriber has a digital certificate installed on their device it can also provide a Zero Sign-on experience since the certificate can be used to authenticate the subscriber instead of a username and password.

One challenge with subscriber digital certificates is installing them on the subscriber's device 12 in a secure manner. One approach is to have the subscriber download and install a certificate manager application that creates a Certificate Signing Request (CSR) and sends it to a certificate issuing server. Before the server can issue the certificate it needs to authenticate the subscriber. If just a username and password is used then the increased authentication strength provided by the certificate is lost. Security is only as strong as its weakest link. A stronger form of subscriber authentication may be used to control issuance of the digital certificates.

The DOCSIS network 34 described above or another type of network (e.g., a cellular network between the access point and the service provider) can be used to support two factor subscriber authentication which is suitable for issuing subscriber certificates. The certificate issuing server (e.g., IdP 40, CMTS 32, or other element with the trust domain 38 of a particular service provider 14) may perform two factor subscriber authentication by doing the following: 1—verifying the subscriber has a valid username and password, such as by requiring the subscriber to perform a single sign-on; and 2—verifying that the CSR request has a source IP address that is associated with the subscriber's authenticated cable modem 30, such as according to the address assurance process described below. If both of these verification steps are successful, the authentication level of the subscriber should be strong enough for the certificate issuing server to issue a certificate to the subscriber's device.

The certificate may contain the subscriber's name or username and the MAC address of the user device, access point, and/or CMTS 32. Including the MAC address may be helpful in preventing the certificate from being shared across devices. Once a certificate is installed on the user device 12 it can be used to automatically authenticate them when they access services or content on the Internet without using a username and password. The CableLabs OLCA specification, the disclosure of which is hereby incorporated by reference in its entirety, defines how cable operators and their partners may use SAML federated single sign-on authentication to enable subscribers to access their video content on line using their cable operator account credentials. Subscribers are normally authenticated with a username and password, but certificates can be used to provide a zero sign-on experience.

When the subscriber connects to the cable operator's site for authentication, on their home network or on an external network, zero sign-on can be accomplished with a mutually authenticated Transport Layer Security (TLS) connection. Certificates can be automatically exchanged between server and client. If certificate validation is successful the server responds with a SAML Assertion redirect command that sends the subscriber back to the service provider for assertion validation and access to the requested content/service. Certificates provide a stronger level of authentication and therefore do not have to be renewed as frequently as the cookie noted above, e.g., not more than once every six months or longer. When a certificate needs to be renewed, say after a one year time period, the certificate manager application connects to the certificate issuing server and installs a new certificate. If the existing certificate has not yet expired, certificate renewal can occur on or off the home

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network since both the valid certificate and the subscriber's username & password can be used for two factor authentication. If the existing certificate has expired certificate renewal may need to occur on the home network to provide two factor authentication for the certificate issuing server as described with initial certificate issuance.

As supported above, the present invention contemplates a zero sign-on process where trust credentials (e.g., a cookie or a certificate) is/are stored on the user device 12 to facilitate automatic, zero sign-on, authentication and access to media services. FIGS. 4a-4b illustrate a flowchart 70 of a method for supporting a zero sign-on process where use of both of the cookie and the certificate are described in accordance with one non-limiting aspect of the present invention. While the use of one or more of the cookie and certificate is described, the present invention fully contemplates facilitating a similar zero sign-on process that relies solely on the processes associated with one of the cookie and the certificate, i.e., for use in environments where use of only the cookie or only the certificate is desirable.

Block 72 relates to a user device connecting to an access point for the first time or before being assigned a cookie or a certification, such as when the user device is initially deployed. The access point, for exemplary non-limiting purposes, is described predominately with respect to being a cable modem or similar edge type device to which a router or other gateway is connected to facilitate wireless communications with one or more user devices. The access point, however, need not be such a device, and particularly, not such a device where the connection thereto is controlled locally by its operator. The access point may be a CMTS or other interface to one of the service provider supported networks.

The exemplary illustration of the user device connecting to a local wireless router that then communicates with the service provider by way of the cable modem and the CMTS is shown as it relates to a particular scenario where the contemplated zero sign-on may be helpful in allowing subscribers to continuously access services while their user device is mobile between trusted and untrusted access points. This is believed to predominately occur with respect to the user devices moving on wireless networks, i.e., those supported with a wireless router. Of course, cellular and/or broadcast networks may not include a router as the access point and instead rely on another device, such as but not limited to a macro, micro, pico, or femto cell. The present invention fully contemplates the zero sign-on process applying to these other types of access points and the corresponding communication mediums.

Block 74 relates to assessing whether the connected access point is trusted. The access point may be considered to be one of trusted and untrusted depending on whether it is within the trust domain of the service provider from which the user device is attempting to access media services. This assessment may require the user device to identify the media services it is attempting to access or to otherwise identifying the service provider from which services are requested. One test of trustworthiness may be based on the IP address assigned to the user device at the time of making the request to access the media services. If the IP address is within a trust domain of IP addresses of the service provider, e.g., within the IP subnet of one of the service provider's trusted CMTSs, the access point may be considered trusted, and if not, untrusted.

Block 72 may be returned to if the access point is untrusted since it may be undesirable to initially provision the user device with a trust credential (i.e., one or both of the

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cookie and/or certificate) through an untrusted access point. Optionally, Block 76 relates to the user being given the option to access the requested media service by entering their username and password in a sign-on operation. The user can then access entitled media services prior to being provisioned with a trust credential if the username and password is verified.

Block 78 relates to the access point being trusted and assessing whether a sign-on operation has been implemented. The sign-on operation may be an optional step where the subscriber decides whether to provision the user device with a cookie and/or a certificate as the trust credential. Optionally, rather than allowing the subscriber to decide, the service provider may require a certain action in that the subscriber may be forced to sign-on and/or the service provider may automatically force cookie provisioning, such as based on subscriber contractual commitments and/or subscriber behavior (e.g., if the subscriber and/or user device is or has been conducting operations considered inappropriate, provisioning of the more secure and long lasting certificate may be prevented).

Blocks 80, 82 relate to provisioning the user device with one or both of the cookie and the certificate. The certificate may require transmission to the access point through in-band signaling or through a proprietary portion of the server provider's trusted domain in order to increase the security of its transmission. The cookie may be transmitted in a similar manner and/or it may be transmitted through out-of-band signaling or over the internet since its security is less of a concern, e.g. the cookie may be transmitted through a Secure Sockets Layer (SSL) pipe. The SSL pipe may be less secure than the in-band signaling since the in-band signaling occurs between known endpoints and the SSL pipe need not.

The foregoing portions of the zero sign-on method may be generally referred to as a provisioning stage in that the user device is provisioned to store one or both of the cookie and certificate. This stage, however, may be optional and avoided in the event the user device has already been provisioned with a trust credential, such as at the time of purchase or installation (preferably installation by an authorized, trusted representative of the service provider). The next portion of the flowchart may be generally referred to as an authentication stage where the user device is authenticated, potentially with a zero sign-on operation, to access media services according to the stored trust credential.

Block 84 relates to determining a request for media services from the user device. This assessment may be the same as the assessment above in Block 72 with respect to connecting the user device to the access point. Block 84 is shown simply to illustrate a beginning of the authentication stage and it may be skipped in the event a sufficient determination has already been made. Block 84 may be triggered upon issuance of the request to access media services, such as if the user device accesses a website to request content from a particular service provider. Other triggering events may also be equivalent to determining a need for authenticating the user devices, such as turning "on" of the user device, entering "send" when dialing a phone call (e.g., to authenticate for cellular access), etc.

Block 86 relates to assessing whether the access point being used by the user device at the time of requesting access to the media services is a trusted access point. The assessment of whether an access point is trusted is sufficiently described herein, and in the event it has already been determined in Block 74, Block 86 may be skipped accordingly. Block 88 is particularly concerned with the trusted/untrusted assessment being used to differentiate whether the

trust credential is needed to facilitate zero sign-on. As described below in more detail, one non-limiting aspect of the present invention contemplates facilitating zero sign-on without reliance on the trust credential, such as based on the trusted state of the access point.

Block **88** relates to the access point being determined to be trusted. This may occur, for example, if the user device is connected to an access point at the home of the subscriber associated with the user device, i.e., the person authenticated to be the principal owner of the user device, and/or that the user device is located at another home or other location where the access point is associated with someone other than the owner of the user device where the corresponding access point is still within the trust domain of the service provider, e.g., if the subscriber takes their user device to a neighbor and connects to the neighbor's access point.

Block **88** also differentiates the next phase of the authentication stage depending on whether the trust credential stored on the user device is a cookie and/or a certificate. In the event both types of trust credentials are stored on the user device, one or the other may be manually selected by the user or automatically by the client application used, such as the certificate since it is likely to be the most secure. Optionally, however, the particular access point and/or media service may dictate the used trust credential, e.g., a certain website may be more conducive to use of the cookie than the certificate.

Block **90** relates to performing a certificate based zero sign-on operation where the user device is authenticated to access the media services associated with the certificate stored thereon. In the event multiple certificates are stored, such as if the user device is used with different subscribers, the user device may be configured to automatically select the certificate without requiring user interaction in order to maintain zero sign-on functionality, such as by selecting the certificate to correspond with the identity of the user as dictated by current and past operating characteristics and performance of the user device, i.e., the client application may be able to identify the user without the user having to input information such that the certificate associated with that user may be automatically selected.

Block **92** relates to performing a cookie based zero sign-on operation where the user device is authenticated to access the media services associated with the cookie stored thereon. In the event multiple cookies are stored, such as if the user device is used with different subscribers, the user device may be configured to automatically select the cookie without requiring user interaction in order to maintain zero sign-on functionality, such as by selecting the cookie to correspond with the identity of the user as dictated by current and past operating characteristics and performance of the user device, i.e., the client application may be able to identify the user without the user having to input information such that the cookie associated with that user may be automatically selected.

Optionally, the cookie may include a relatively small amount of data or otherwise not be configured to specifically identify the user and/or the entitlements of the user with respect to the media services. This may be done in order to ameliorate processing demands and/or backend operations required to categorize cookies for large numbers of users. Instead, the cookie may be used simply to indicate a trusted entity such that the user device, even though it is not specifically verified, is assumed to be trustworthy because it has the cookie.

Optionally, the use of the cookie in Block **92** may be omitted if the service provider is willing to accept the fact

that the user device is connected to a trusted access point as sufficient proof of its trustworthiness. This scenario may be used to facilitate whole home access to the same types of content, such as in a traditional television transmission scenario where all televisions in a home can access the same content to any device allowed to connect to the access point. Additional options may include at least partially relying on the cookie in order to grant access to certain media services, such as is common with the need for a STB or DVR to access certain television signaling not otherwise available to all televisions within the same home.

In the event the cookie is not used, or the used cookie is not self-identifying and/or it is desirable to avoid performing other operations to identify the user device and/or user, such as by matching MAC address or performing some other mapping operation, the entitlements/media permission authenticated to the user device may instead be those associated with the access point. Since the service provider is likely to have already referenced a billing address and other media permission to the access point, extending those same permissions to the user device may be done relatively simply, i.e., without having to perform extra MAC or IP address matching that would not otherwise have to be performed.

Block **94** relates to refreshing the cookie if used and/or stored on the user device. The refresh may be used to extend the validity of the cookie beyond that specified in its last provisioning. Rather than disrupting service when a cookie is being used on a relatively frequent and trusted basis, the cookie may be automatically refreshed to maintain zero sign-on continuity. This can be helpful in maintaining zero sign-on capabilities for users that frequently access the same websites and/or media services.

Returning to Block **86**, in the event the access point is deemed to be untrusted, Block **96** is reached to make an assessment of which trust credential is to be used. Block **96** may be reached, for example, if instead of taking the user device to a neighbor's home network, the user takes the user device to a Wi-Fi hotspot outside of the service provider's trust domain. The trust credential selected in Block **96** may be determined in a process equivalent to the process in Block **88**. If the certificate is used, the process of Block **90** is used to facilitate access to the media services. Since the provisioning of the certificate is done in a more restrictive manner than the cookie, it may be acceptable to the service provider to disregard the untrustworthiness of the access point in light of the relatively trustworthy certificate.

Block **98** relates to assessing whether the cookie is valid in the event the user device does not include a certificate or the stored certificate is otherwise not used as part of the authentication process. The cookie may be valid for a limited period of time, as described above. Block **100** relates to requiring the user to complete a sign-on operation if the cookie is invalid so that access can be granted to the media service. Block **102** relates to performing zero sign-on in the event the cookie is valid, such as in a manner equivalent to Block **92**. Optionally, Block **102** may require use of the cookie to execute the zero sign-on, as opposed to Block **92** which permits zero sign-on without reliance on the cookie due to the access point being trusted. Since the access point is untrusted, Block **94** may be bypassed in order to force connection of the user device to a trusted access point before permitting refresh.

As noted, the IP source addresses may be used to support zero sign-on capabilities. The present invention contemplates implementing an address assurance process to insure the IP addresses are verified and not spoofed or otherwise

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compromised. Optionally, the zero sign-on contemplated by the present invention may rely on a DOCSIS cable modem (access point) MAC address and the client application's IP source address.

The cable modem MAC address spoofing or cloning can be prevented by the following: enforcing DOCSIS BPI+. BPI+ uses digital certificates to authenticate the CM, including MAC address validation (DOCSIS 3.0, the disclosure of which is hereby incorporated by reference in its entirety provides a BPI+ Enforce feature); and MAC address domain validation wherein the cable modem provisioning system only allows a cable modem to operate in an authorized CMTS MAC address domain such that if a hacker tries to clone the MAC address and use it in a different MAC address domain the provisioning system will block it (note: The CMTS blocks duplicate MAC addresses in the same domain).

The IP Address spoofing can be prevented by the following: router configuration (device connected cable modem/ access point) may be required to prevent traffic from outside of the MSO's trusted domain (e.g., outside network (Internet)) to have source IP addresses within the MSO's client subnet; and source address verification where client applications on the MSO's network can only use IP addresses assigned by the MSO's provisioning system (DOCSIS 3.0 provides this feature). While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

What is claimed is:

1. A method for authenticating a device for zero sign-on (ZSO) access to a service available through one or more access points, the method comprising:

(i-a) ensuring an Internet Protocol (IP) subnet address of the device matches an IP subnet address of a first access point of the one or more access points;

(i-b) determining a Media Access Control (MAC) address for the first access point as being associated with the device when a mapping of device IP addresses and MAC addresses performed upstream of the first access point maps a device IP address of the device to the MAC address of the first access point;

(ii) identifying the first access point and the device as falling within a trusted domain when the subnet address of the device matches the subnet address of the first access point in step (i-a) and the MAC address of the first access point is determined to be associated with the device in step (i-b); and

(iii) facilitating transport of a trust credential to the device when the first access point and the device fall within the trusted domain, thereby enabling the device to use the trust credential to facilitate ZSO access to the service, wherein the mappings of the device IP addresses and the MAC addresses is generated prior to a credential request being issued from the device to the first access point, the credential request being used to request transport of the trust credential, to the device, and

wherein the credential request includes the MAC address of the first access point.

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2. The method of claim 1 further comprising:

determining a service request received from the device through the first access point, the service request being issued to request access to the service through the first access point; and

facilitating ZSO access to the service through the first access point when the trust credential has been transported to the device through the first access point.

3. The method of claim 1 further comprising identifying the first access point or the device as being a subscriber of the service.

4. The method of claim 1 further comprising:

generating the mapping according to notifications issued from a dynamic host configuration protocol (DHCP) server, the notifications each individually associating one of the device IP addresses with one of the MAC addresses.

5. The method of claim 1 further comprising:

(iv) determining the MAC address to be unassociated with the device when the mapping fails to map a device IP address of the device to the MAC address of the first access point.

6. The method of claim 5 further comprising:

(v) identifying the first access point and the device as failing to fall within the trusted domain when the MAC address of the first access point is determined to be unassociated with the device in step (iv); and

(vi) denying transport of the trust credential through the first access point to the device when the first access point and the device fail to fall within the trusted domain.

7. The method of claim 6 further comprising:

when the first access point and the device fail to fall within the trusted domain, instructing the device to request the trust credential from another of the one or more access points or to perform a sign-on process to access the service.

8. The method of claim 1 further comprising:

determining a service request received from the device through a second access point of the one or more access points, the service request being issued to request access to the service through the second access point.

9. The method of claim 8 further comprising:

instructing the device to transport a user identifier input while connected to the second access point to facilitate non-ZSO access through the second access point to the service when the trust credential is either not received from the device through the second access point or the trust credential is expired.

10. The method of claim 8 further comprising:

facilitating ZSO access to the service through the second access point when the trust credential is received from the device through the second access point.

11. The method of claim 10 further comprising:

limiting the ZSO access according to entitlements associated with the received trust credential.

12. The method of claim 8, wherein the second access point is untrusted.

13. The method of claim 8, wherein the first access point and the second access point facilitate access to different access networks.

14. The method of claim 1 further comprising:

polling a plurality of termination stations operable with the one or more access points to provide the service thereto, including determining whether any of the termination stations have a duplicate of the MAC address.

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15. The method of claim **14** further comprising:
denying transport of the trust credential to the device
when the duplicate is determined.

16. The method of claim **15** further comprising:
instructing the device to perform a sign-on process to
access the service when transport of the trust credential
has been denied. 5

17. A method for authenticating a device for zero sign-on
(ZSO) access to a service available through one or more
access points, the method comprising:

- (i) determining a credential request issued from the device
to a first access point of the one or more access points,
the credential request being used to request transport of
a trust credential to the device, the trust credential being
sufficient for authenticating ZSO access to the service; 10
- (ii) identifying a Media Access Control (MAC) address
for the first access point added to the credential request
by the first access point; 15
- (iii-a) ensuring an Internet Protocol (IP) subnet address of
the device matches an IP subnet address of the first
access point; 20
- (iii-b) determining the MAC address for the first access
point as being associated with the device when a
mapping of device IP addresses and MAC addresses
performed upstream of the first access point maps a
device IP address of the device to the MAC address of
the first access point; 25

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(iv) identifying the first access point and the device as
falling within a trusted domain when the subnet address
of the device matches the subnet address of the first
access point in step (iii-a) and the MAC address of the
first access point is determined to be associated with the
device in step (iii-b); and

(v) facilitating transport of the trust credential to the
device when the first access point and the device fall
within the trusted domain, thereby enabling the device
to use the trust credential to facilitate ZSO access to the
service,

wherein the mapping of the device IP addresses and the
MAC addresses is generated prior to a credential
request being issued from the device to the first access
point, the credential request being used to request
transport of the trust credential to the device.

18. The method of claim **17** further comprising:

determining a service request received from the device
through the first access point, the service request being
issued to request access to the service through the first
access point; and

facilitating ZSO access to the service through the first
access point when the trust credential has been trans-
ported to the device through the first access point.

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